
Algorithm 1 Ribozyme-mediated circularization scoring of a candidate IRES

Input: candidate IRES s ; permuted group-I self-splicing ribozyme halves $R_{3'}$ (upstream) and $R_{5'}$ (downstream, carries the IGS); reporter ORF

Output: circularization score $r(s)$; catalytic splice site j^*

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1:  $\mathcal{L} \leftarrow \text{HAIRPINLOOPS}(s)$  {MFE-folded, accessible loop + stable stem}
2:  $r \leftarrow 0, j^* \leftarrow \emptyset$ 
3: for each loop  $\ell \in \mathcal{L}$ , each candidate site  $j \in \ell$  do
4:   if  $j$  fails the splice-site sequence rule then
5:     continue
6:   end if
7:    $R_{5'}^{(j)} \leftarrow \text{REWRITEIGS}(R_{5'}; s, j)$  {pair IGS to the site upstream of  $j$ }
8:    $S \leftarrow R_{3'} + s[j:] + \text{ORF} + s[:j] + R_{5'}^{(j)}$  {PIE construct}
9:   if ribozyme halves in  $S$  fail to fold into an active conformation then
10:    continue
11:  end if
12:   $P \leftarrow \text{PAIRPROB}(S)$  {ViennaRNA partition function}
13:  if IGS-target pairing at the splice junction is weak then
14:    continue
15:  end if
16:   $\alpha \leftarrow \text{STABILITYSCORE}(P; R_{3'}, R_{5'})$ 
17:  if  $\alpha > r$  then
18:     $r \leftarrow \alpha, j^* \leftarrow j$ 
19:  end if
20: end for
21: return  $(r, j^*)$ 
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